

1. What is Retrieval-Augmented Generation (RAG) ?

Retrieval-Augmented Generation combines the best of both worlds of the retrieval system and a generative model. The RAG approach involves first retrieving relevant documents or snippets of data from a large corpus to subsequently use such information in informing and augmenting responses generated by a neural network. This way, the method is very helpful in systems that demand a rich, contextually accurate generation of texts, such as chatbots, content creation, and question-answering systems.

2. How does RAG differ from traditional generative models ?

As compared to traditional generative models, which simply rely on the patterns learned during their pre-training, RAG models augment the outputs with data from various data sources such as video, images, texts, articles,... etc., and generate a response using an LLM for retrieving the data for a user query. This retrieval step allows RAG models to produce responses that are not only contextually relevant but also updated and informed by the most recent information available. This hybrid approach leads to better accuracy and relevance of the generated content compared to purely generative model

3. What challenges are associated with implementing RAG models?

RAG models are more difficult to implement than vanilla LLMs because they demand a comprehensive and well-maintained knowledge base from which to retrieve information. Moreover, it requires good computational power and optimization for the retrieval and generation components to work seamlessly in real-time applications. For the retrieved information, its accuracy and relevance are key in the process, considering that errors in the retrieval can directly result in errors in the generated content.

